A Short Empirical Study of Hawkes Process Models for Earthquake Occurrences

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Abstract

We compare a homogeneous Poisson process with marked and unmarked exponential and power-law Hawkes processes on an earthquake catalogue from the USGS Earthquake Hazards Program. Models are fitted by maximum likelihood and evaluated via the time-rescaling theorem. Kolmogorov–Smirnov and chi-squared tests on the rescaled residuals consistently favour the marked exponential Hawkes model.

1 Introduction

Point process models are a standard tool for studying the temporal structure of earthquake occurrences and quantifying aftershock activity. The simplest benchmark is a homogeneous Poisson process, which assumes independent exponentially distributed inter-event times. However, observed catalogues typically exhibit strong temporal clustering, with bursts of activity following large events, which motivates the use of self-exciting Hawkes processes.

In this short note we compare five models on an earthquake catalogue from the USGS Earthquake Hazards Program: a homogeneous Poisson process, unmarked and marked exponential and power-law Hawkes processes. Our goal is to identify a parsimonious specification that captures the observed clustering and magnitude dependence of aftershock activity, as assessed by residual diagnostics under the time-rescaling theorem.

2 Data

We use an earthquake catalogue retrieved from the United States Geological Survey (USGS) Earthquake Hazards Program database,[1] which provides event times, locations and magnitudes for earthquakes worldwide. We select earthquakes occurring between 01/01/2025 and 04/05/2026 within the United States of America. The data set contains $N = 2961$ events over an observation window of approximately 449 days.

For the purpose of this study we ignore spatial information and model only the temporal occurrence times, treating the catalogue as a one-dimensional point process on $[t_0, T]$. We centre and scale the times so that $t_0 = 0$ and keep magnitudes in their original units. A lower magnitude cut-off M_0 is imposed to ensure approximate completeness of the catalogue.

3 Models and methods

3.1 Point process models

Let $(N_t)_{t\geq 0}$ denote the counting process associated with the earthquake times $t_1 < \cdots < t_N$ on $[0, T]$. All models are specified through their conditional intensity $\lambda(t \mid \mathcal{F}_t)$, where $\mathcal{F}_t$ is the history up to time t.

[1] https://earthquake.usgs.gov/earthquakes/search/

The homogeneous Poisson model assumes a constant intensity $\lambda(t) \equiv \lambda > 0$, so that the inter-event times are i.i.d. Exp(λ). In contrast, Hawkes processes have the form

$$\lambda(t) = \mu + \sum_{t_i < t} g(t - t_i, M_i),$$

where $\mu > 0$ is a background rate and g is a triggering kernel. We consider four Hawkes specifications:

- Exponential kernel (unmarked): $g(u, M) = \alpha e^{-\beta u} \mathbf{1}_{\{u>0\}}$.
- Power-law kernel (unmarked): $g(u, M) = K(c + u)^{-p} \mathbf{1}_{\{u>0\}}$.
- Marked exponential kernel: $g(u, M) = \alpha e^{\gamma(M - M_0)} e^{-\beta u} \mathbf{1}_{\{u>0\}}$.
- Marked power-law kernel: $g(u, M) = K e^{\gamma(M - M_0)} (c + u)^{-p} \mathbf{1}_{\{u>0\}}$.

All parameters are constrained to ensure positivity and stability (e.g. $\alpha/\beta < 1$ in the exponential case).

3.2 Maximum likelihood estimation

For a given intensity model $\lambda_\theta(t)$ with parameter vector θ, the log-likelihood of the observed times is

$$\ell(\theta) = \sum_{i=1}^{N} \log \lambda_\theta(t_i) - \int_0^T \lambda_\theta(t)\, \mathrm{d}t.$$

For the homogeneous Poisson model the MLE is $\hat{\lambda} = N/(T - t_0)$. For all Hawkes models we evaluate the log-likelihood numerically and maximise it using the `scipy.minimize` routine with a quasi-Newton method and simple box constraints on the parameters.

3.3 Residual analysis via time rescaling

To assess absolute goodness of fit we use Ogata's time-rescaling theorem. Given a fitted model with intensity $\lambda_{\hat{\theta}}$, we compute the integrated intensity $\Lambda_{\hat{\theta}}(t) = \int_0^t \lambda_{\hat{\theta}}(s)\, \mathrm{d}s$ and define transformed times $\tau_i = \Lambda_{\hat{\theta}}(t_i)$. Under correct specification, the rescaled gaps $\Delta\tau_i = \tau_i - \tau_{i-1}$ should be i.i.d. Exp(1). We therefore apply Kolmogorov–Smirnov and chi-squared tests against the Exp(1) law and compare the empirical CDFs and densities of the $\Delta\tau_i$ to their theoretical exponential counterparts. This allows us to check not only relative fit, but also whether a given model can be regarded as an adequate description of the data in absolute terms.

4 Results

4.1 Empirical evidence of temporal clustering

Figure 1 compares the empirical CDF of the inter-event times $\Delta t_i = t_i - t_{i-1}$ with the Exp($\hat{\lambda}$) CDF, where $\hat{\lambda} = N/(T - t_0)$. The empirical curve lies above the Poisson benchmark near the origin and below it for intermediate gaps, indicating an excess of very short inter-event times and a deficit of medium gaps. This pattern is characteristic of aftershock clustering and already suggests that a homogeneous Poisson process is inadequate.

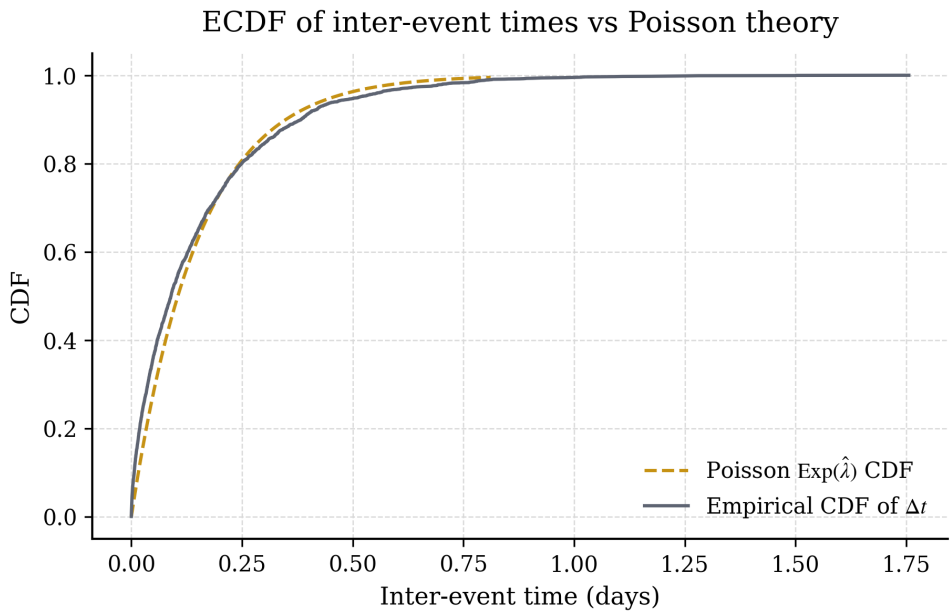

Figure 1: Empirical CDF of Δt_i (solid) against the Exp($\hat{\lambda}$) CDF (dashed).

Table 1: MLE parameter estimates.

Model	Estimates
Poisson	$\hat{\lambda} = 6.59$
Exponential Hawkes	$(\hat{\mu}, \hat{\alpha}, \hat{\beta}) = (5.47, 9.16, 53.98)$
Power-law Hawkes	$(\hat{\mu}, \hat{K}, \hat{c}, \hat{p}) = (3.30, 0.05, 0.10, 1.50)$
Marked exp. Hawkes	$(\hat{\mu}, \hat{\alpha}, \hat{\beta}, \hat{\gamma}) = (5.52, 5.02, 55.85, 1.12)$
Marked power-law Hawkes	$(\hat{\mu}, \hat{K}, \hat{c}, \hat{p}, \hat{\gamma}) = (4.38, 0.067, 0.55, 5.00, 1.09)$

4.2 Fitted models

Table 1 reports the MLE parameter estimates for all five models. The Hawkes fits indicate substantial self-excitation, and the marked models yield $\hat{\gamma} > 0$, confirming a positive dependence of triggering strength on magnitude. Figure 2 shows the fitted conditional intensities for the three main models: the Poisson intensity is constant, whereas both Hawkes intensities are spiky, with the marked variant producing sharper peaks after large earthquakes.

4.3 Residual diagnostics

We apply Ogata's time-rescaling theorem: under correct specification, the rescaled gaps $\Delta\tau_i = \Lambda_{\hat{\theta}}(t_i) - \Lambda_{\hat{\theta}}(t_{i-1})$ should be i.i.d. Exp(1). The Poisson and power-law Hawkes models are both strongly rejected by KS and chi-squared tests ($p < 10^{-19}$). The exponential Hawkes model performs substantially better (KS statistic 0.022, $p \approx 0.12$), and the marked exponential Hawkes model provides the best agreement (KS statistic 0.017, $p \approx 0.39$; chi-squared $p \approx 0.07$). Figures 3 and 4 confirm this visually: the marked exponential Hawkes residuals are nearly indistinguishable from the Exp(1) benchmark, while all other models show clear discrepancies.

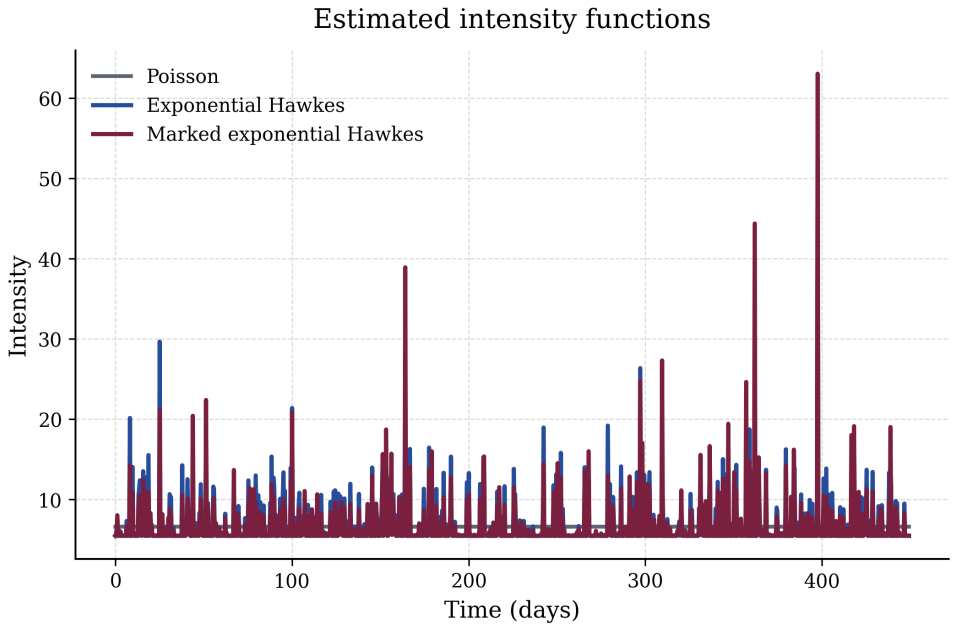

Figure 2: Estimated conditional intensity functions for the Poisson, exponential Hawkes and marked exponential Hawkes models.

5 Conclusion

Using an earthquake catalogue from the USGS database, we compared a homogeneous Poisson process with several Hawkes process specifications, including marked and unmarked exponential and power-law kernels. Simple diagnostics on the raw inter-event times reveal pronounced temporal clustering that is incompatible with a constant-rate Poisson model. Residual analyses under the time-rescaling theorem, based on Kolmogorov–Smirnov and chi-squared tests, point to the marked exponential Hawkes process as the most adequate representation of the data: it captures both short-term aftershock clustering and the positive dependence of triggering intensity on earthquake magnitude, yielding rescaled inter-arrivals that are approximately Exp(1).

Several limitations remain. The analysis is purely temporal and ignores the spatial distribution of events; a space–time Hawkes process would likely provide a more faithful description of seismicity in the region. The background rate μ is assumed constant throughout the observation window, which may not hold over long periods or across distinct seismic sequences.

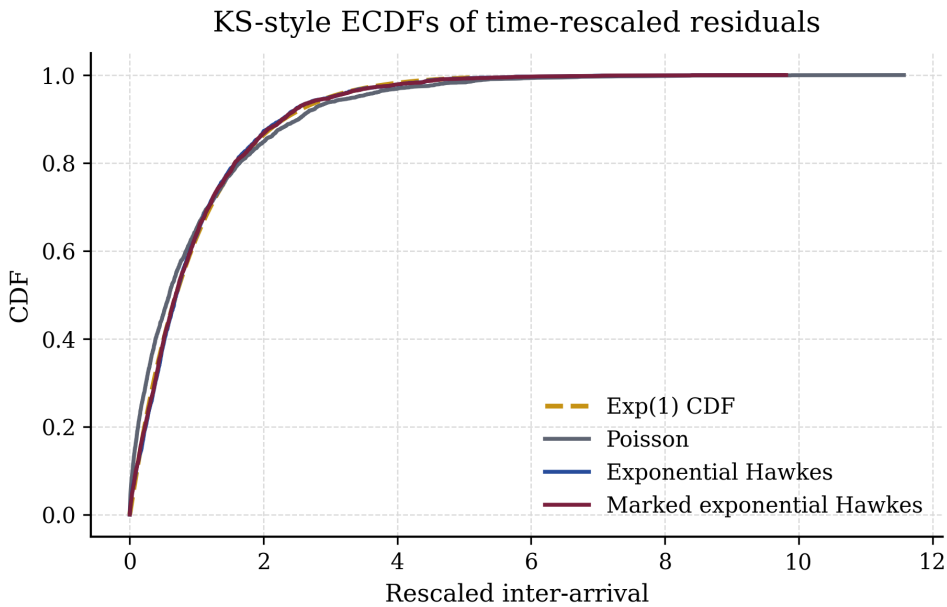

Figure 3: Empirical CDFs of the rescaled inter-arrival times against the Exp(1) CDF (dashed).

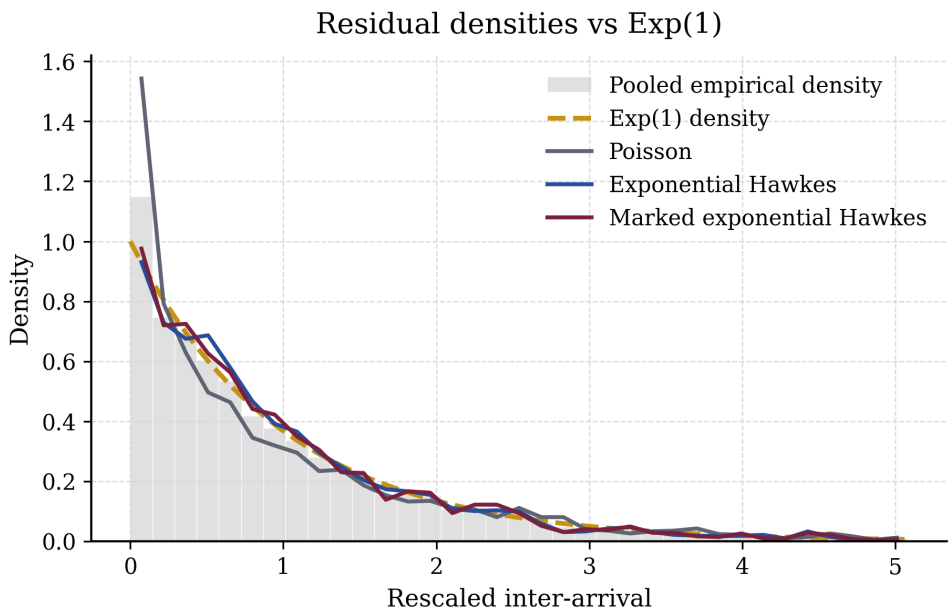

Figure 4: Residual densities against the Exp(1) density (dashed).